AWS re/Start Program
Mini Project Assignment

Project Task Description:
Objective: Launch and configure an EC2 instance to host a simple web server.
Tasks:

1. **Launch an EC2 Instance:**
   - Create a new EC2 instance using the AWS Management Console.
   - Select an appropriate Amazon Machine Image (AMI) (e.g., Amazon Linux 2).
   - Choose an instance type (e.g., t2.micro) that is eligible for the free tier.
2. **Configure Security Groups**:
   - Create or select a security group that allows HTTP (port 80) and SSH (port 22) access.
   - Ensure the security group allows inbound traffic from My IP for SSH and Anywhere for HTTP.
3. **Connect to Your Instance:**
   - Use an SSH client (e.g., PuTTY, Terminal) to connect to your EC2 instance.
   - Ensure you have the key pair file (.ppk or .pem) needed for SSH access.
4. **Install and Configure a Web Server:**
   - Update the instance's package manager (e.g., sudo yum update for Amazon Linux).
   - Install a web server (e.g., Apache: sudo yum install httpd).
   - Start the web server and ensure it runs on boot (e.g., sudo systemctl start httpd and sudo systemctl enable httpd).
5. **Deploy a Simple Web Page:**
   - Create a simple HTML file (e.g., index.html) and place it in the web server's root directory (e.g., /var/www/html for Apache).
   - Verify that the web page is accessible via the EC2 instance's public IP address.
6. **Submission:**
   - Provide the public IP address of your EC2 instance with the web server running.
   - Submit a screenshot of the web page hosted on your EC2 instance.
   - Optionally, document any additional configurations or features you implemented.

Resources:
AWS Documentation: Refer to AWS documentation on EC2 instance creation, security groups, and web server setup for detailed guidance.

Walkthrough Video: Link

Tips for Implementation of Project Task
Below is a step-by-step guide for connecting to the EC2 instance and installing the web server. Use it as a guide to complete the task.

Step-by-Step Guide

Step 1: Launch an EC2 Instance

1. **Log in to AWS Management Console:**
    - Go to the AWS Management Console.
    - Navigate to the EC2 Dashboard.
2. **Launch Instance:**
    - Click on the "Launch Instance" button.
    - Select an Amazon Machine Image (AMI), such as "Amazon Linux 2 AMI (HVM), SSD Volume Type".
    - Choose an instance type (e.g., t2.micro, which is eligible for the free tier).
    - Click "Next: Configure Instance Details".
3. **Configure Instance:**
    - Leave the default settings.
    - Click "Next: Add Storage".
4. **Add Storage:**
    - Leave the default settings.
    - Click "Next: Add Tags".
5. **Configure Security Group:**
    - Create a new security group with the following rules:
        - Type: SSH, Protocol: TCP, Port Range: 22, Source: My IP
        - Type: HTTP, Protocol: TCP, Port Range: 80, Source: Anywhere
    - Click "Review and Launch".
6. **Review and Launch:**
    - Click "Launch".
    - Select an existing key pair or create a new key pair.
    - Download the key pair file (.ppk/.pem) and keep it secure.
    - Click "Launch Instances".
7. **View Instances:**
    - Click "View Instances" to see your running instance.

Step 2: Connect to Your EC2 Instance

1. **Locate Your Instance:**
   - On the EC2 Dashboard, select your instance.
   - Note the "Public IPv4 address" of your instance.
2. **Connect Using SSH:**

   For Linux/macOS (Using Terminal):
   - Find the key pair used when the instance was launched (Example: my-key.pem)
     - If you do not have this file, SSH access is not possible without recovery steps
   - Set correct permissions on the key file (Linux / macOS) by running:
     - chmod 400 my-key.pem
     - This prevents SSH from rejecting the key for being publicly accessible.
   - Get the instance Public IP or DNS
   - From EC2 instance details, copy one of:
     - Public IPv4 address (e.g., 3.121.xxx.xxx)
     - Public IPv4 DNS (e.g., ec2-3-121-xxx-xxx.compute.amazonaws.com)
   - SSH into the instance
     - Use the correct default username based on the AMI:
       - Amazon Linux - ec2-user
       - Ubunt - ubuntu
       - RHEL - ec2-user
       - CentOS - centos
     - Run the SSH command using the syntax:
       - ssh -i <keypair.pem> ec2-user@<PUBLIC_IP>
       - example: ssh -i my-key.pem ubuntu@3.123.100.111
   - Accept the host key (first-time only).
     - When prompted:
       - Are you sure you want to continue connecting (yes/no)?
     - Type:
       - yes

   For Windows 8 below (Using PuTTY):

   - Open PuTTY and enter the public IP address of your instance.
   - Under "Connection -> SSH -> Auth", browse and select the .ppk file.
   - Click "Open" to connect.

Step 3: Install and Configure a Web Server

1. **Update the Package Manager:**
   sudo yum update -y

2. **Install Apache Web Server:**
   sudo yum install httpd -y

3. **Start the Apache Web Server:**
   sudo systemctl start httpd

4. **Enable Apache to Start on Boot:**
   sudo systemctl enable httpd

Step 4: Deploy a Simple Web Page

1. **Create an HTML File:**
   Hint: Open a Nano text editor and enter the text in below. Save as (e.g,. index.html)
   echo "<html><h1>My First EC2 Project</h1> Created by: *Fullname*. Email: *Your email*. </html>" | sudo tee /var/www/html/index.html

   Note:
   Replace **"Fullname"** with your full name.
   Replace **"Your email"** with the email you registered for the re/Start program.
2. **Verify the Web Page:**
   - Open a web browser.
   - Enter the public IP address of your EC2 instance.
   - You should see the simple HTML page you created.

Step 5: Terminate the Instance (After Completion)

1. **Terminate Your Instance:**
   - Go to the EC2 Dashboard.
   - Select your instance.
   - Click "Actions" -> "Instance State" -> "Terminate".

Deliverables

Deliverable	Evidence Required
1. EC2 Instance Launch	1. Screenshot of the EC2 Console showing the Instance ID and Name in a running state. *File name: EC2_01_InstanceRunning_FullName*
2. Security Group Configuration	2. Screenshot of security group rules showing: - SSH (Port 22) → Source: My IP - HTTP (Port 80) → Source: Anywhere *File name: EC2_02_SecurityGroupRules_FullName*
3. Successful SSH Connection	3. Screenshot of a successful SSH login to the EC2 instance *File name: EC2_03_SSH_Login_FullName*
4. Web Server Installation & Configuration	4. Screenshot of running Apache service (**must show active (running)**) *Run sudo systemctl status httpd* *File name: EC2_04_ApacheRunning_FullName*
5. Deployment of Web Page	5. Screenshot of **index.html** in /var/www/html/ *File name: EC2_05_IndexHTML_FullName* 6. Screenshot of the output of curl localhost on SSH terminal (**must return web page content**) *File name: EC2_06_CurlLocalhost_FullName*
6. Public Web Page Verification	7. Screenshot of the web page loaded in a browser, showing the **public IP address in the browser's address bar** *File name: EC2_07_PublicAccess_FullName*
7. Termination & Cleanup	8. Screenshot of the **terminated EC2 instance** after verification *File name: EC2_08_InstanceTerminated_FullName*

Refer to the table and submit your screenshots as evidence for each deliverable to this form via the link below.
Link to the form: Click here to access the submission form.

Please observe the deadline and time.